amount of voltage (Ir) has dropped across the internal resistance *r*.

Then $V = \xi - Ir$

$$Ir = \xi - V \qquad (2.36)$$

Dividing equation (2.36) by equation (2.35), we get

$$\frac{Ir}{IR} = \frac{\xi - V}{V}$$

$$r = \left[\frac{\xi - V}{V}\right] R \qquad (2.37)$$

Since ξ, V and R are known, internal resistance r can be determined. We can also find the total current that flows in the circuit.

Due to this internal resistance, the power delivered to the circuit is not equal to power rating mentioned in the battery. For a battery of emf ξ, with an internal resistance r, the power delivered to the circuit of resistance R is given by

$P = I\xi = I\,(V + Ir)$ (from equation 2.36)

Here V is the voltage drop across the resistance R and it is equal to IR.

Therefore, P = I (IR +Ir)

$$P = I^2 R + I^2 r \qquad (2.38)$$

Here I^2 r is the power delivered to the internal resistance and I^2 R is the power delivered to the electrical device (here it is the resistance R). For a good battery, the internal resistance r is very small, then $I^2 r << I^2$ R and almost entire power is delivered to the resistance.

EXAMPLE 2.17

A battery has an emf of 12 V and connected to a resistor of 3 Ω. The current in the circuit is 3.93 A. Calculate (a) terminal voltage and the internal resistance of the battery (b) power delivered by the battery and power delivered to the resistor

Solution

The given values I = 3.93 A, ξ = 12 V, R = 3 Ω

(a) The terminal voltage of the battery is equal to voltage drop across the resistor

V = IR = 3.93 × 3 = 11.79 V

The internal resistance of the battery,

$$r = \left[\frac{\xi - V}{V}\right] R = \left[\frac{12 - 11.79}{11.79}\right] \times 3 = 0.05\ \Omega$$

(b) The power delivered by the battery P = Iξ = 3.93 × 12 = 47.1 W

The power delivered to the resistor = I^2 R = 46.3 W

The remaining power = (47.1 – 46.3) P = 0.772 W is delivered to the internal resistance and cannot be used to do useful work. (it is equal to $I^2 r$).

2.4.3 Cells in series

Several cells can be connected to form a battery. In series connection, the negative terminal of one cell is connected to the positive terminal of the second cell, the negative terminal of second cell is connected to the positive terminal of the third cell and so on. The free positive terminal of the first cell and the free negative terminal of the last cell become the terminals of the battery.

Suppose n cells, each of emf ξ volts and internal resistance r ohms are connected in series with an external resistance R as shown in Figure 2.21